PHILIP KING

Antikythera Engine Demo: beardedphil.com
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SOFTWARE ENGINEER | AI SYSTEMS ARCHITECT

CAREER SUMMARY

Senior Software Engineer with 10+ years of experience spanning frontend architecture, distributed systems, and AI-augmented development platforms. Architect of Antikythera Engine, an AI-driven orchestration system designed to bring determinism, auditability, and reliability to software development workflows. Proven track record of modernizing legacy systems, delivering major performance gains, and designing scalable frameworks that improve both developer velocity and system correctness.

SKILLS

- React (Advanced)
- TypeScript
- JavaScript (ES6+)
- Custom Component Architecture
- Frontend Performance Optimization
- Microfrontend Architecture
- Vite
- Code Quality Enforcement (ESLint, Prettier)
- Git & GitHub Workflow
- RESTful APIs
- Agile Delivery Leadership
- AI-Augmented Development Systems
- Distributed Systems (NATS, Microservices)
- LLM Orchestration & Prompt Design
- Retrieval-Augmented Generation (RAG)
- System Observability & Execution Tracing

PROFESSIONAL EXPERIENCE

Antikythera Engine **August 2025 – Current**
Founder / Lead Architect (Independent)

Architect and creator of an AI-driven software development system designed to orchestrate end-to-end engineering workflows. Focused on building deterministic, auditable agent pipelines that transform how software is planned, implemented, tested, and documented.

- Introduced a deterministic alternative to traditional agent-based development workflows, addressing reliability, verification, and trust challenges in AI-assisted software engineering.
- Architected an AI-driven development orchestration platform enabling autonomous planning, implementation, validation, and documentation through a deterministic 8-stage execution pipeline.

- Designed a runtime-supervised agent system that enforces correctness through structured validation layers (build, test, coverage, documentation sync), eliminating reliance on LLM self-verification.
- Implemented a multi-service architecture using Go microservices, NATS-based messaging, and a React/TypeScript UI with real-time SSE streaming for execution visibility.
- Developed a working-set and retrieval system combining embeddings, symbol indexing, and import graph analysis to deliver high-signal context to LLM agents.
- Eliminated unbounded LLM tool loops by introducing stage-constrained execution, significantly improving reliability, observability, and cost predictability.
- Designed a full execution trace and evidence system, enabling auditability of AI decision-making and reproducibility of system behavior across runs.
- Delivered a live demo pipeline showcasing end-to-end autonomous feature implementation, including planning, code edits, test execution, and documentation updates.
- Demonstrates a scalable approach to AI-assisted software development that reduces engineering overhead while increasing system reliability and traceability.

Centene **April 2024 – January 2026**

Senior Application Development Engineer

Lead UI Developer for an AI-powered contract processing platform. Although officially a contractor, operates as the de facto lead for the UI team, overseeing development work by three full-time Centene engineers (including a Lead App Dev Engineer). Direct all UI planning, architecture, and implementation, working with significant autonomy to modernize the codebase, improve team velocity, and deliver major performance gains.

- **Team Leadership & Mentorship**: Supervise a rotating group of three internal developers. Regularly onboard new engineers, review code, assign tickets, and answer technical questions. Serve as the go-to resource for all frontend guidance.
- **Architect of Modern UI Infrastructure**: Designed and implemented a fully custom React table component featuring pagination, expandable nested rows, and dynamic contract rendering—all optimized for performance and scalability.
- **Massive Performance Win**: Reduced main page load time from **15 seconds to 1 second**, improving responsiveness and setting a new benchmark for app performance.
- **Proactive Code Quality Champion**: Introduced and enforced code standards using **ESLint** and **Prettier**, then refactored all the legacy UI to conform to these standards—improving consistency, readability, and maintainability.
- **Owner of UI Delivery**: Solely responsible for sprint planning, backlog grooming, and work breakdown for all frontend work. Backend is managed separately, but all UI scoping, estimates, and execution originates from me.
- **Initiative-Driven Contributor**: Routinely identify and execute high-value improvements beyond assigned work, including core refactors, architectural upgrades, and process enhancements.

Bayer **March 2021 – February 2024**
Software Engineer

Highly skilled software developer contributing to marketing and consumer-facing applications. Thrives under tight deadlines and cross-functional collaboration. Instrumental in establishing a groundbreaking application that shifted the company towards crop science R&D and global expansion. Developed an advanced reporting suite supporting a multinational corporation with diverse crops using innovative data structures.

- Provided mentorship and technical expertise in UI, React, and JavaScript to a team of 7 engineers during the migration from Angular to React, contributing significantly to a seamless transition that was completed on target and on budget.
- Spearheaded the implementation of polygon map overlays, overcoming technical complexities as an individual contributor while driving collaboration with the data asset team.
- Designed a complex multi-layer pivot table that effectively manages extensive datasets and intricate features, all while adhering to stringent performance standards.

AT&T **January 2018 – February 2020**
Software Engineer

Developed and maintained business process software for this global telecom leader. Updated and supplied troubleshooting for frontend, backend, and database solutions, all using Agile methodologies. Became proficient in Angular, JavaScript, Python, Django REST Framework, SQL, and Git.

- Supported large SQL queries (1000+ lines) used to power a project metrics dashboard for C-level management, including updating documentation.
- Built major updates to an internal job posting and hiring application, using Angular, Java, and SQL.
- Identified additional value-add opportunities and delivered by expanding scope of work beyond development to include resolving technical debt and improving UI/UX.

EDUCATION

Washington University in St. Louis **May 2017**
B.S. in Mathematics and Managerial Economics
(Final Honors – 3.94 Cumulative G.P.A)

PROFESSIONAL TRAINING

Udemy

Modern React with Redux [2024 Update]	2024
Modern React with Redux [2020 Update]	2020
The Ultimate MySQL Bootcamp	2019
The Complete React Developer Course	2017